
LAADAN-AC: A Lagrangian Admissibility-Aware Actor-Critic for Safe Offline Decision-Making

From ICU-Sepsis Benchmarking toward Adaptive Robot Control

Riya Basak

University of Hertfordshire

BSc Computer Science (Hons) with Artificial Intelligence

Research Note

Abstract

This research note presents LAADAN-AC, a Lagrangian Admissibility-Aware Deep Action-Nudging Actor-Critic developed from an Intelligent Adaptive Systems coursework project. The immediate benchmark environment is ICU-Sepsis, a fixed tabular Markov decision process for reinforcement-learning evaluation in sepsis-treatment-style decision making, not a clinical deployment environment. The wider research aim is to investigate whether the same safety structure can later support adaptive robot-control systems, especially where unsafe, unsupported or behaviourally implausible actions must be avoided. LAADAN-AC combines actor-critic learning with admissibility masking, twin reward critics, a cost critic, Lagrangian cost control, expert-policy regularisation and smoothness-aware action penalties. Across the supplied multi-seed results, LAADAN-AC achieved strong survival/return while maintaining zero inadmissibility and the highest expert argmax agreement among the learned policies. The results support the central interpretation that, in offline safety-critical reinforcement learning, raw return alone is insufficient: policy quality must also be judged by support, safety and expert consistency. The future direction is to connect this safety-aware policy layer with JEPA-inspired latent world models for safer adaptive robot behaviour.

Keywords: offline reinforcement learning; constrained Markov decision process; actor-critic learning; Lagrangian safety; admissibility masking; ICU-Sepsis; social robotics; world models; JEPA.

1. Introduction

Adaptive agents should not only optimise reward. In safety-critical environments, they must also avoid actions that are unsafe, unsupported or behaviourally implausible. This is the central motivation behind LAADAN-AC. The model was first implemented on the ICU-Sepsis benchmark, where the agent chooses among 25 discretised treatment actions from a fixed patient-state representation. However, the long-term research vision is broader: to explore whether the same constrained actor-critic structure can become a safer adaptive-control framework for real-world and social robotics.

The design follows the constrained Markov decision process view, where a policy maximises expected return while satisfying a cost constraint (Altman, 1999). In reinforcement learning, this principle also appears in constrained policy optimisation, where policy improvement is coupled with explicit safety or cost constraints (Achiam et al., 2017). LAADAN-AC adopts this constrained-learning motivation, but combines it with project-specific mechanisms for admissibility-aware action selection and expert-guided policy shaping.

The benchmark setting also matters. ICU-Sepsis is useful because it provides a standardised Markov decision process for algorithm evaluation, but it should not be interpreted as a system for drawing direct medical

treatment conclusions (Choudhary et al., 2024). This distinction is important: the project evaluates safe offline reinforcement-learning behaviour in a benchmark MDP, not a real clinical recommender.

2. Background and Related Work

Offline reinforcement learning learns from fixed data rather than from live environment interaction. This makes it attractive for safety-sensitive domains, but it also creates a major risk: a learned policy may exploit estimation errors and select actions that appear valuable but are poorly supported by the dataset (Kumar et al., 2020; Levine et al., 2020). In healthcare-style settings, reinforcement-learning evaluation therefore needs careful attention to support, uncertainty and safety rather than reward alone (Gottesman et al., 2019).

Conservative Q-Learning (CQL) addresses part of this problem by discouraging over-optimistic value estimates for out-of-distribution actions (Kumar et al., 2020). Behaviour cloning (BC) addresses a different part by imitating the expert policy, often producing stable but limited policies. Actor-critic methods, including soft actor-critic-style entropy regularisation, can improve policy optimisation but may become unsafe if unconstrained (Haarnoja et al., 2018). LAADAN-AC combines these ideas into a safety-guided actor-critic: conservative value learning reduces unsupported value inflation, while direct admissibility masking and expert regularisation shape the final action-selection process.

The longer-term motivation comes from adaptive robotics and world-model learning. Safe human-robot interaction requires systems that avoid harmful or implausible actions when operating near people (Lasota et al., 2017). Robot learning from demonstration also motivates expert-guided behaviour shaping, especially when purely reward-driven learning is insufficient (Ravichandar et al., 2020). Recent JEPA-style world-model work suggests that latent predictive models can support efficient planning from compact representations (LeCun, 2022; Maes et al., 2026). LAADAN-AC is therefore positioned as a possible future policy-safety layer above a learned world model, rather than as a complete robotics system in its current form.

3. Benchmark Setting

The current implementation uses the ICU-Sepsis benchmark, a tabular MDP derived from real medical data for reinforcement-learning research. In this project configuration, the benchmark exposes 716 states, 25 actions, 47-dimensional state-centre features, a finite horizon of 20, terminal death and survival states, reward-by-next-state structure, an expert policy and an admissible-action mask. The released benchmark is used as a fixed offline environment for training and exact finite-horizon evaluation.

The task is not to perform treatment in the real world. The task is to learn a policy that chooses one of the benchmark’s 25 discrete actions from each state while balancing return, admissibility and expert consistency. This makes the benchmark suitable for studying the core offline-RL problem: whether a learned policy can achieve good return without drifting into unsupported action regions.

4. Proposed Method: LAADAN-AC Architecture

LAADAN-AC was designed around the main weakness of offline reinforcement learning: a policy can look strong on return while selecting actions that are poorly supported by the fixed dataset. This is especially risky in healthcare-style benchmarks, where a learned policy should not drift into clinically implausible treatment recommendations (Gottesman et al., 2019; Levine et al., 2020).

The method addresses this risk in five ways. First, the admissibility mask prevents the actor from selecting unsupported actions. Second, conservative critic regularisation reduces overvaluing weakly supported actions, following the motivation of CQL (Kumar et al., 2020). Third, expert-policy regularisation keeps the policy close to the benchmark expert-safe policy. Fourth, the smoothness penalty discourages abrupt action changes,

following the wider safety concern that sepsis decision-support systems should avoid unsafe or clinically implausible recommendations (Jia et al., 2020, 2021). Fifth, the Lagrangian cost term treats inadmissible behaviour as a constraint: if expected cost rises above the budget, the multiplier increases and adds stronger safety pressure. This follows constrained reinforcement-learning work, where the policy optimises reward while respecting safety constraints (Achiam et al., 2017).

The intended trade-off is therefore clear: LAADAN-AC may sacrifice some raw return compared with an unrestricted actor-critic, but it should produce a safer and more expert-consistent policy.

4.1. Architecture Diagram

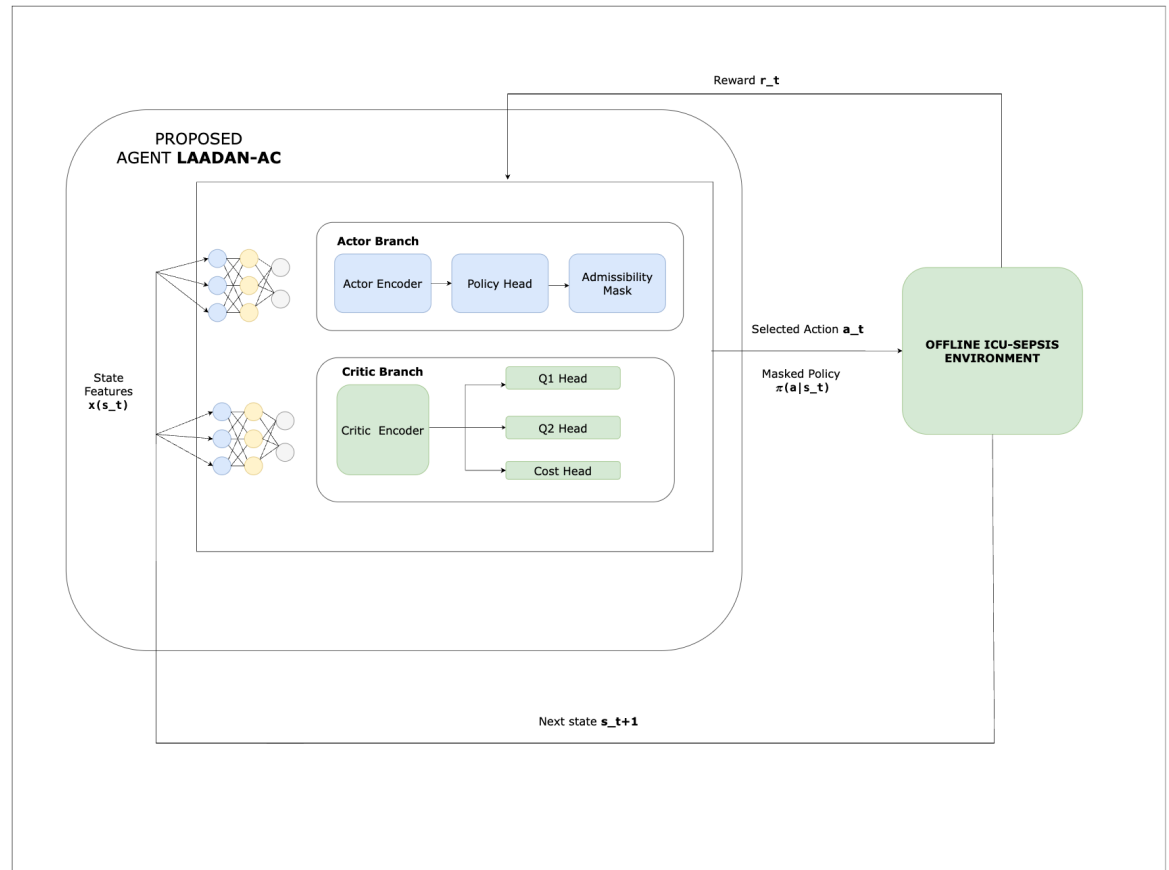


Figure 1: LAADAN-AC architecture and offline reinforcement-learning loop. The input is the benchmark state representation $x(s_t)$, implemented as released 47-dimensional standardised state-centre features. The actor branch produces action logits for the 25 treatment actions, after which the admissibility mask removes unsupported actions and produces the masked policy $\pi(a | s_t)$. The selected action is evaluated inside the fixed ICU-Sepsis benchmark MDP. In parallel, the critic branch estimates reward value through twin critics Q_1 and Q_2 and estimates safety cost through a separate cost head. The actor is therefore guided by reward, admissibility, conservative critic regularisation, expert-policy regularisation, smoothness regularisation and the Lagrangian cost term.

Figure 1 should be drawn as an actor-critic reinforcement-learning architecture rather than as a generic software-system diagram. The key message is that the actor chooses actions, while the critic evaluates both return and safety cost. The admissibility mask sits directly between action logits and the final policy, because this is the main mechanism that prevents unsupported actions from being selected.

4.2. Notation

Table 1: Main notation used in the LAADAN-AC formulation.

Symbol	Meaning
s	Current state.
a	Candidate action.
$\pi_\theta(a s)$	Actor policy parameterised by θ .
$z_\theta(s, a)$	Actor logit for action a in state s .
$m(s, a)$	Admissibility mask; 1 for allowed actions, 0 for blocked actions.
Q_1, Q_2	Twin reward critics.
$Q(s, a)$	Conservative reward estimate, defined as $\min(Q_1, Q_2)(s, a)$.
$C_\psi(s, a)$	Cost critic estimating unsafe or inadmissible cost.
π_E	Expert-safe policy used for regularisation.
$S(s, a)$	Smoothness or action-deviation cost used in this project.
λ	Lagrange multiplier controlling cost pressure.
d	Allowed cost budget.
α, β, μ	Weights for entropy, expert guidance and smoothness.

For state s and action a , the actor first produces logits $z_\theta(s, a)$. LAADAN-AC then applies an admissibility mask:

$$\pi_\theta(a | s) = \frac{\exp(z_\theta(s, a))m(s, a)}{\sum_{a'} \exp(z_\theta(s, a'))m(s, a')}. \quad (1)$$

Here, $m(s, a) = 1$ means that action a is admissible in state s , while $m(s, a) = 0$ means that the action is blocked. This masking mechanism is a project-specific LAADAN-AC design choice and provides the direct action-level safety filter used by the final policy.

5. Constrained Objective

The safety problem is written as a constrained optimisation problem:

$$\max_{\pi_\theta} J_R(\pi_\theta) \quad \text{subject to} \quad J_C(\pi_\theta) \leq d. \quad (2)$$

Here, $J_R(\pi_\theta)$ is the expected return and $J_C(\pi_\theta)$ is the expected unsafe cost. In the current implementation, $d = 0$, so the policy is encouraged toward zero unsafe cost.

Following constrained reinforcement learning, the constraint can be relaxed using a non-negative Lagrange multiplier λ (Altman, 1999; Achiam et al., 2017):

$$\mathcal{L}(\pi_\theta, \lambda) = J_R(\pi_\theta) - \lambda (J_C(\pi_\theta) - d). \quad (3)$$

In LAADAN-AC, this appears in the actor objective as an adaptive cost penalty.

6. Actor and Critic Learning

The actor minimises the following proposed LAADAN-AC loss:

$$\mathcal{L}_{\text{actor}} = \mathbb{E}_s \left[-\mathbb{E}_{a \sim \pi_\theta} [Q(s, a)] - \alpha \mathcal{H}(\pi_\theta) + \lambda \mathbb{E}_{a \sim \pi_\theta} [C_\psi(s, a)] + \beta D_{\text{KL}}(\pi_E \| \pi_\theta) + \mu \mathbb{E}_{a \sim \pi_\theta} [S(s, a)] \right]. \quad (4)$$

The first term maximises reward value. The entropy term encourages a less brittle policy and is motivated by maximum-entropy actor-critic learning (Haarnoja et al., 2018). The cost term penalises unsafe predicted cost. The expert KL term keeps the policy close to the expert-safe policy. The smoothness term discourages unusually rough or extreme action choices. The expert KL and smoothness terms are project-specific LAADAN-AC additions.

The Lagrange multiplier is updated by projected ascent:

$$\lambda \leftarrow \max(0, \lambda + \eta_\lambda(\bar{C} - d)), \quad (5)$$

where η_λ is the Lagrange learning rate and $\bar{C}$ is the current mean expected cost. If the expected cost exceeds the budget, λ increases, making unsafe actions more expensive in later updates. This is the mathematical reason LAADAN-AC is described as Lagrangian.

The reward and cost critics are fitted to finite-horizon Bellman targets computed from the benchmark transition and reward functions. The reward critics estimate long-term return, while the cost critic estimates long-term unsafe cost. A conservative critic regulariser is also used to discourage unsupported value inflation, following the motivation of CQL (Kumar et al., 2020).

The important distinction is that the hard zero-inadmissibility behaviour mainly comes from the admissibility mask, while the Lagrangian term adds adaptive pressure during learning. LAADAN-AC is therefore not claiming that the Lagrangian term alone guarantees safety. Instead, safety is encouraged through a combination of masking, cost estimation, expert guidance and constrained optimisation.

7. Algorithm

Listing 1: LAADAN-AC training procedure.

```

Input:
  Fixed ICU-Sepsis benchmark MDP with transition table P, reward table R,
  immediate-cost table, smoothness-cost table, state features X,
  expert-safe policy pi_E, admissible-action mask M, actor network,
  critic networks, target networks, Lagrange multiplier lambda = 0,
  discount factor gamma, target update rate tau, conservative weight alpha,
  expert-KL weight eta, smoothness weight mu, entropy coefficient beta,
  cost budget d, Lagrange learning rate rho, number of epochs E,
  and evaluation interval K.

for epoch = 1, 2, ..., E do
  1. Use the target actor to build a masked target policy:
     pi_target(a|s) = masked_softmax(actor_target(X), M)

  2. Estimate target reward values using the smaller target critic:
     min_Q_target = min(Q1_target, Q2_target)

  3. Build the Bellman reward target.

  4. Build the Bellman cost target.

  5. Update the two reward critics by minimising Bellman error.

  6. Update the cost critic by minimising cost Bellman error.

  7. Add conservative critic regularisation to reduce over-valuing

```

```

    weakly supported actions.

    8. Use the current actor to build the current masked policy:
        pi(a|s) = masked_softmax(actor(X), M)

    9. Compute the actor loss using reward maximisation, admissibility
        awareness, cost penalty, expert-policy regularisation, smoothness
        regularisation and entropy regularisation.

    10. Update the actor parameters.

    11. Update the Lagrange multiplier:
        lambda <- max(0, lambda + rho(mean_expected_cost - d))

    12. Soft-update the target networks.

    13. Every K epochs:
        build the masked greedy policy,
        evaluate it using exact finite-horizon policy evaluation,
        and save the checkpoint if survival improves.
end for

```

Algorithm 1 summarises the training loop. The important implementation idea is that the policy is not trained only to maximise Bellman return. It is repeatedly shaped by the admissibility mask, the conservative critic, the expert-safe policy, the smoothness penalty and the Lagrangian cost signal.

8. Experimental Design

The evaluation compares LAADAN-AC against three baselines: behaviour cloning, CQL and vanilla offline actor-critic. The comparison focuses on five metrics. Survival/return measures task performance. Inadmissibility measures whether the selected action violates the benchmark admissibility mask. Expert argmax match measures whether the learned policy chooses the same top action as the expert-safe policy. KL to expert measures full-distribution divergence from the expert policy. Mean action deviation measures how far the chosen action is from the expert-style action index.

This metric set is important because a policy with the highest return is not necessarily the safest policy. For offline safety-critical reinforcement learning, a useful learned policy should combine strong return with low inadmissibility and plausible expert-aligned behaviour.

9. Results

Table 2: Multi-seed training/evaluation results, reported as mean $\pm$ 95% confidence interval.

Model	Survival / Return	Inadmissibility	Expert argmax match	KL to expert	Mean action deviation
BC	0.7862 $\pm$ 0.0016	0.0000 $\pm$ 0.0000	0.9409 $\pm$ 0.0109	3.9599 $\pm$ 0.0134	1.3459 $\pm$ 0.0268
CQL	0.7839 $\pm$ 0.0017	0.0008 $\pm$ 0.0004	0.8949 $\pm$ 0.0044	4.0506 $\pm$ 0.0276	1.5324 $\pm$ 0.0732
VOAC	0.8017 $\pm$ 0.0028	0.4179 $\pm$ 0.0532	0.1304 $\pm$ 0.0935	15.5813 $\pm$ 1.6694	10.3319 $\pm$ 1.7637
LAADAN-AC	0.7931 $\pm$ 0.0007	0.0000 $\pm$ 0.0000	0.9525 $\pm$ 0.0043	3.9531 $\pm$ 0.0036	1.4358 $\pm$ 0.0139

Table 2 shows the main return-safety trade-off. VOAC achieved the highest survival, but also had very high inadmissibility and very poor expert agreement. LAADAN-AC did not maximise raw survival as aggressively as VOAC, but achieved the strongest balanced result: high survival, zero inadmissibility and the highest expert argmax match.

Table 3: Final selected checkpoints and testing verification.

Model	Selected seed	Final survival / return	Inadmissibility	Expert match	KL to expert	Testing result
BC	43	0.7881	0.0000	0.9464	3.9526	Passed
CQL	43	0.7852	0.0009	0.8972	4.0381	Passed
VOAC	44	0.8045	0.4075	0.0924	16.4088	Passed
LAADAN-AC	43	0.7941	0.0000	0.9548	3.9520	Passed

Table 3 confirms reproducibility of the final selected models. The selected checkpoints were reloaded and tested again, and the saved-against-recomputed checks passed. This supports the claim that the final results are tied to stored models and saved configuration rather than one-off reporting.

Table 4: Soft-policy safety check.

Model	Soft survival	Soft inadmissibility	Interpretation
BC	0.7785	0.0075	Mostly safe, but the full probability distribution still placed small mass on inadmissible actions.
CQL	N/A	N/A	CQL was evaluated as a greedy Q-value policy in this project, so no soft-policy metrics were produced.
VOAC	0.7762	0.8534	Unsafe under the full learned distribution, showing that its high greedy survival was not robust.
LAADAN-AC	0.7915	0.0000	Safe under both greedy and soft-policy evaluation.

Table 4 is important because it evaluates the full learned action distribution where that distribution exists. BC, VOAC and LAADAN-AC produce policy logits, so their soft policies can be evaluated directly. CQL was evaluated as a deterministic greedy policy from Q-values, so it is reported as N/A rather than forcing an artificial softmax over Q-values. VOAC looked strong under greedy action selection, but its soft policy placed very high probability on inadmissible actions. LAADAN-AC remained safe in both greedy and soft forms, supporting the use of masked action selection and safety-aware regularisation.

Table 5: Convergence summary.

Model	First epoch reaching 95% of best survival	Interpretation
BC	10	Survival stabilised early, but policy quality still depended on expert alignment.
CQL	10	Reward stabilised early, while conservative value learning kept inadmissibility low.
VOAC	10	Reached high survival early, but did not converge to a safe policy.
LAADAN-AC	10	Reached strong survival early while maintaining zero inadmissibility.

Evaluation was recorded every 10 epochs, so epoch 10 is the earliest measurable convergence point. This result shows that survival alone converged quickly, but it was not enough to judge policy quality. Safety and alignment metrics were still needed to distinguish the methods.

10. Interpretation of Plots and Convergence

The overall performance comparison shows the main return-safety trade-off. VOAC achieved the highest survival because it was the least restricted actor-critic model and could directly optimise the Bellman return objective. However, this higher return came with high inadmissibility and weak expert agreement, suggesting that the unrestricted actor-critic policy relied more on poorly supported action regions. LAADAN-AC achieved

slightly lower survival, but with zero inadmissibility and the strongest expert match. This was expected from the design: LAADAN-AC was built to trade a small amount of raw reward for safer and more supported action selection.

The benchmark reference comparison gives context for the learned policies. The optimal reference remains higher, so the benchmark was not fully solved. This is reasonable because the optimal reference is computed directly from the known MDP, while the neural models learn through function approximation and regularised objectives. The gap to the optimal reference does not indicate a design fault in LAADAN-AC or the baselines. The optimal policy is computed directly from the known transition and reward tables of the benchmark MDP, so it represents an ideal dynamic-programming reference with full access to the environment model. In contrast, BC, CQL, VOAC and LAADAN-AC are learned neural policies trained with finite optimisation, 47-dimensional state-centre features, regularisation and model-specific constraints.

The survival-during-training curve stabilised early. Using the 95%-of-best-survival rule, all methods reached this level by the first evaluation point at epoch 10. Because evaluation was recorded every 10 epochs, epoch 10 is the earliest measurable convergence point. This does not mean all models were equally good after epoch 10. It means survival alone was not sensitive enough to separate the methods, so safety and alignment metrics were needed.

The inadmissible-action curve gives the clearest evidence for the project aim. VOAC kept a high inadmissible-action rate because its actor was not restricted by the benchmark support mask. LAADAN-AC stayed at zero inadmissibility because masking, cost modelling and the Lagrangian penalty directly changed the action-selection process. This confirms that the safety mechanisms affected behaviour rather than only adding extra losses in the code.

The action-preference heatmap supports the numerical alignment results. BC and LAADAN-AC produced action patterns closer to the expert policy, while VOAC selected noticeably different actions. This explains why VOAC had high survival but very low expert argmax match, while LAADAN-AC had slightly lower survival but much stronger expert consistency.

The sensitivity plots show that LAADAN-AC was not dependent on one arbitrary setting. Conservative alpha had the clearest effect: lower values slightly improved survival while inadmissibility stayed at zero. This suggests that too much conservatism can reduce return, but the admissibility mask still protects safety. Expert-KL and smoothness weights caused smaller changes in the tested range, suggesting that the main safety effect came from the mask and cost-aware design, while expert and smoothness terms refined alignment.

11. Discussion

The results are consistent with the theory of offline reinforcement learning. VOAC directly optimised expected return through Bellman critic targets and actor updates, so it was expected to achieve the highest survival. However, because it had no admissibility mask, no expert regularisation and no cost constraint, it also selected many unsupported actions. This is the central offline RL risk: a policy can exploit value-estimation errors in fixed data and appear strong on reward while becoming unsafe or poorly supported (Kumar et al., 2020; Levine et al., 2020).

CQL reduced this failure mode because its conservative loss discourages over-optimistic Q-values. This explains why CQL had much lower inadmissibility than VOAC. However, CQL still did not match LAADAN-AC's safety and alignment because it controlled value estimation, not the final actor policy through direct masking and expert-guided regularisation. Briefly, CQL made the critic more cautious, while LAADAN-AC made both the critic and the action-selection process safety-aware.

BC behaved as expected for an imitation baseline. It stayed close to the expert policy and achieved zero greedy inadmissibility because it learned from expert supervision rather than freely maximising long-horizon return. Its weakness is also clear: copying the expert gives a stable and safe baseline, but it gives limited scope to improve the return-safety trade-off beyond the demonstrated policy.

LAADAN-AC produced the strongest balanced result in the supplied tables. It did not beat VOAC on raw survival, but it achieved zero inadmissibility, the highest expert argmax match and strong soft-policy safety. This is the most important interpretation of the experiment: the best policy in this benchmark is not simply the one with the highest survival score. For offline safety-critical reinforcement learning, the policy also needs to remain inside supported action regions and stay close to plausible expert behaviour.

This directly relates to adaptive-systems theory. The project uses an MDP, Bellman value learning, actor-critic policy learning, function approximation, exact finite-horizon evaluation and the reward-against-safety trade-off. Unlike online RL, there was no live epsilon-style exploration. The main exploration issue appeared as a dataset-support problem: the model could exploit the fixed MDP by choosing actions that looked valuable but were weakly supported by the offline data. LAADAN-AC reduced this risk by combining admissibility-aware action selection with conservative, expert-guided and cost-aware learning.

12. Future Research Vision: Toward World-Model-Based Robot Control

The current ICU-Sepsis implementation is a benchmark prototype, not a clinical deployment system and not yet a robot-control algorithm. The future research direction is to adapt LAADAN-AC beyond a tabular MDP into safer adaptive robot control. This is motivated by the idea that autonomous agents should learn predictive world models, reason in latent representation space and use those predictions to guide action selection (LeCun, 2022).

A future JEPA-inspired LAADAN-AC would replace the fixed benchmark state with a learned latent world model. In a social robotics setting, the robot would use this world model to predict likely consequences of actions before acting. The LAADAN-AC policy layer would then apply admissibility constraints, Lagrangian cost control, expert guidance and smoothness penalties to select safer actions.

Recent JEPA-based work such as LeWorldModel suggests that joint-embedding predictive architectures can learn compact latent world models from visual inputs for prediction and planning (Maes et al., 2026). This makes LAADAN-AC a possible bridge between safe offline reinforcement learning and world-model-based adaptive robotics. However, this remains a research direction rather than an implemented claim in the current project.

13. Limitations

This proposal should be read as an early-stage research note. The current implementation is evaluated on a benchmark MDP and should not be interpreted as a real clinical or robotics deployment system. Further work would need additional component-level ablations, broader benchmarks, external validation and direct testing in simulated robotics environments before any real-world claims could be made.

A second limitation is that the safety claim is benchmark-specific. Zero inadmissibility means zero violation of the released admissibility mask, not a universal proof of real-world safety. A third limitation is that the future world-model direction is currently conceptual. Connecting LAADAN-AC to visual JEPA representations would require a separate implementation, environment, training pipeline, and evaluation protocol.

14. Conclusion

LAADAN-AC was designed to test whether a constrained actor-critic policy can preserve strong return while avoiding unsupported actions in an offline benchmark. The supplied results show that LAADAN-AC did not maximise raw survival as strongly as VOAC, but it achieved a better safety-alignment profile: zero inadmissibility, strong survival, the highest expert argmax match and robust soft-policy safety. The central contribution is therefore not only a higher reward score; it is the proposed combination of admissibility masking, conservative value learning, expert guidance, smoothness regularisation and Lagrangian cost control. This

makes LAADAN-AC a useful benchmark prototype for studying safer adaptive decision-making, with a clear future path toward world-model-based robot control.

References

- Achiam, J., Held, D., Tamar, A. and Abbeel, P. (2017) ‘Constrained policy optimization’, in *Proceedings of the 34th International Conference on Machine Learning (ICML 2017)*, Proceedings of Machine Learning Research, 70, pp. 22–31. Available at: <https://doi.org/10.48550/arXiv.1705.10528>.
- Altman, E. (1999) *Constrained Markov Decision Processes*. Boca Raton, FL: Chapman & Hall/CRC. Available at: <https://doi.org/10.1201/9781315140223>.
- Choudhary, K., Gupta, D. and Thomas, P.S. (2024) ‘ICU-Sepsis: a benchmark MDP built from real medical data’, *Reinforcement Learning Conference (RLC 2024)*. Available at: <https://doi.org/10.48550/arXiv.2406.05646>.
- García, J. and Fernández, F. (2015) ‘A comprehensive survey on safe reinforcement learning’, *Journal of Machine Learning Research*, 16, pp. 1437–1480. Available at: <https://www.jmlr.org/papers/v16/garcia15a.html>.
- Gottesman, O., Johansson, F., Komorowski, M., Faisal, A., Sontag, D., Doshi-Velez, F. and Celi, L.A. (2019) ‘Guidelines for reinforcement learning in healthcare’, *Nature Medicine*, 25(1), pp. 16–18. Available at: <https://doi.org/10.1038/s41591-018-0310-5>.
- Haarnoja, T., Zhou, A., Abbeel, P. and Levine, S. (2018) ‘Soft actor-critic: off-policy maximum entropy deep reinforcement learning with a stochastic actor’, in *Proceedings of the 35th International Conference on Machine Learning (ICML 2018)*, Proceedings of Machine Learning Research, 80, pp. 1861–1870. Available at: <https://doi.org/10.48550/arXiv.1801.01290>.
- Harris, C.R., Millman, K.J., van der Walt, S.J., Gommers, R., Virtanen, P., Cournapeau, D., Wieser, E., Taylor, J., Berg, S., Smith, N.J., Kern, R., Picus, M., Hoyer, S., van Kerkwijk, M.H., Brett, M., Haldane, A., del Río, J.F., Wiebe, M., Peterson, P., Gérard-Marchant, P., Sheppard, K., Reddy, T., Weckesser, W., Abbasi, H., Gohlke, C. and Oliphant, T.E. (2020) ‘Array programming with NumPy’, *Nature*, 585, pp. 357–362. Available at: <https://doi.org/10.1038/s41586-020-2649-2>.
- Hunter, J.D. (2007) ‘Matplotlib: a 2D graphics environment’, *Computing in Science & Engineering*, 9(3), pp. 90–95. Available at: <https://doi.org/10.1109/MCSE.2007.55>.
- Hussein, A., Gaber, M.M., Elyan, E. and Jayne, C. (2017) ‘Imitation learning: a survey of learning methods’, *ACM Computing Surveys*, 50(2), Article 21. Available at: <https://doi.org/10.1145/3054912>.
- JGraph Ltd. (2026) *diagrams.net*. Available at: <https://www.diagrams.net/> (Accessed: 22 May 2026).
- Jia, Y., Burden, J., Lawton, T. and Habli, I. (2020) ‘Safe reinforcement learning for sepsis treatment’, in *2020 IEEE International Conference on Healthcare Informatics (ICHI)*. IEEE, pp. 1–7. Available at: <https://doi.org/10.1109/ICHI48887.2020.9374367>.
- Jia, Y., Lawton, T., Burden, J., McDermid, J. and Habli, I. (2021) ‘Safety-driven design of machine learning for sepsis treatment’, *Journal of Biomedical Informatics*, 117, 103762. Available at: <https://doi.org/10.1016/j.jbi.2021.103762>.
- Kober, J., Bagnell, J.A. and Peters, J. (2013) ‘Reinforcement learning in robotics: a survey’, *The International Journal of Robotics Research*, 32(11), pp. 1238–1274. Available at: <https://doi.org/10.1177/0278364913495721>.
- Komorowski, M., Celi, L.A., Badawi, O., Gordon, A.C. and Faisal, A.A. (2018) ‘The Artificial Intelligence Clinician learns optimal treatment strategies for sepsis in intensive care’, *Nature Medicine*, 24(11), pp. 1716–1720. Available at: <https://doi.org/10.1038/s41591-018-0213-5>.

- Kumar, A., Zhou, A., Tucker, G. and Levine, S. (2020) ‘Conservative Q-learning for offline reinforcement learning’, in *Advances in Neural Information Processing Systems*, 33, pp. 1179–1191. Available at: <https://doi.org/10.48550/arXiv.2006.04779>.
- Lasota, P.A., Fong, T. and Shah, J.A. (2017) ‘A survey of methods for safe human-robot interaction’, *Foundations and Trends in Robotics*, 5(4), pp. 261–349. Available at: <https://doi.org/10.1561/23000000052>.
- LeCun, Y. (2022) ‘A path towards autonomous machine intelligence’, *OpenReview position paper*. Available at: <https://openreview.net/pdf?id=BZ5a1r-kVsf> (Accessed: 22 May 2026).
- Levine, S., Kumar, A., Tucker, G. and Fu, J. (2020) ‘Offline reinforcement learning: tutorial, review, and perspectives on open problems’, *arXiv preprint arXiv:2005.01643*. Available at: <https://doi.org/10.48550/arXiv.2005.01643>.
- Lillicrap, T.P., Hunt, J.J., Pritzel, A., Heess, N., Erez, T., Tassa, Y., Silver, D. and Wierstra, D. (2016) ‘Continuous control with deep reinforcement learning’, in *Proceedings of the 4th International Conference on Learning Representations (ICLR 2016)*. Available at: <https://doi.org/10.48550/arXiv.1509.02971>.
- Maes, L., Le Lidec, Q., Scieur, D., LeCun, Y. and Balestrieri, R. (2026) ‘LeWorldModel: stable end-to-end joint-embedding predictive architecture from pixels’, *arXiv preprint arXiv:2603.19312*. Available at: <https://doi.org/10.48550/arXiv.2603.19312>.
- Paszke, A., Gross, S., Massa, F., Lerer, A., Bradbury, J., Chanan, G., Killeen, T., Lin, Z., Gimelshein, N., Antiga, L., Desmaison, A., Köpf, A., Yang, E., DeVito, Z., Raison, M., Tejani, A., Chilamkurthy, S., Steiner, B., Fang, L., Bai, J. and Chintala, S. (2019) ‘PyTorch: an imperative style, high-performance deep learning library’, in Wallach, H., Larochelle, H., Beygelzimer, A., d’Alché-Buc, F., Fox, E. and Garnett, R. (eds.) *Advances in Neural Information Processing Systems*, 32, pp. 8024–8035. Available at: <https://doi.org/10.48550/arXiv.1912.01703>.
- Python Software Foundation (2026) *The Python Standard Library*. Available at: <https://docs.python.org/3/library/> (Accessed: 22 May 2026).
- Ravichandar, H., Polydoros, A.S., Chernova, S. and Billard, A. (2020) ‘Recent advances in robot learning from demonstration’, *Annual Review of Control, Robotics, and Autonomous Systems*, 3, pp. 297–330. Available at: <https://doi.org/10.1146/annurev-control-100819-063206>.
- Schulman, J., Wolski, F., Dhariwal, P., Radford, A. and Klimov, O. (2017) ‘Proximal policy optimization algorithms’, *arXiv preprint arXiv:1707.06347*. Available at: <https://doi.org/10.48550/arXiv.1707.06347>.
- Sutton, R.S. and Barto, A.G. (2018) *Reinforcement Learning: An Introduction*. 2nd edn. Cambridge, MA: MIT Press. Available at: <http://incompleteideas.net/book/the-book-2nd.html>.